Amir Nabiyev

Computer Science Student | AI Researcher

amirn2203@gmail.com (347) 935-4619 New York, NY AmirNabiyev30 amirnabiyev

Summary

Computer science student and AI researcher at The City College of New York, with experience in automated theorem proving, AI agents, fraud detection, and computer science instruction.

Education

Bachelor of Science **The City College of New York (CCNY), CUNY**, Computer Science — New York, NY, May 2027

Experience

Air Force Research Laboratory, Visiting Faculty Fellowship Program — Rome, NY, June 2026 – present, 6 months

- Designed a hybrid AI theorem-proving architecture integrating large language models and formal verification to improve mathematical proof generation and correctness.

The City College of New York, AI Agents Research Assistant — New York, NY, Mar 2026 – present, 9 months

- Collaborating with Dr. Saptarashmi Bandyopadhyay to explore research directions with AI agents.
- Designing and prototyping AI agent workflows for finance portfolio optimization.

The City College of New York, Computer Science Tutor — New York, NY, Jan 2024 – present, 2 years 11 months

- Delivered personalized instruction in computer science fundamentals, including C++ and Python.
- Led workshops on creating web-based projects and solving data structures and algorithms problems.
- Assisted students in developing debugging strategies and reasoning skills to improve code quality.

Projects

Automated Theorem Proving System

Python, LangGraph, LangChain, Lean 4

- Rebuilt the Goedel-Architect framework for automated theorem proving, using LLM agents to generate structured proof blueprints and iteratively refine proof strategies.
- Integrated Lean formal verification into the proof pipeline to validate generated mathematical statements and proofs.
- Extended the architecture with a new proof-search skill and evaluated its ability to improve automated reasoning on research-level mathematical problems.

Elder Fraud Protection

React, Python, JavaScript, Flask, OpenAI

- Developed an AI-powered fraud detection platform that analyzes messages and emails for scam indicators and generates risk assessments with explanations.
- Built a Chrome extension that extracts email content from Gmail and Outlook and integrates with a backend analysis pipeline for automated screening.
- Implemented authenticated user workflows and backend infrastructure with MongoDB and Clerk, with deployment support for Vercel and Render.

nanoGPT
Python, PyTorch

- Built a transformer-based language model from the ground up, implementing key ideas for text generation.
- Improved model performance through experimentation with training strategies to enhance output quality.

Skills

Languages: Python, JavaScript, TypeScript, C++, SQL

Frameworks: React, PyTorch, TensorFlow, FastAPI, Tailwind CSS, Next.js, LangChain, LangGraph, Ollama

Developer Tools: Git, GitHub Actions, MLflow, SQLite, Docker, AWS, PostgreSQL, MongoDB